

Ethernet Security

Module 24

Duration: 34:12

Video: <https://www.youtube.com/watch?v=uaDXhq0k5sc>

Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Prof Yogesh Chaba, Department of Computer Engineering and IT, Guru Jambheshwar University of Science & Technology, Hisar

Auto-generated YouTube transcript for personal study. Spoken wording belongs to the original lecture. Follow the video link above; this file is not an official CEC/SWAYAM publication.

0:00 [Music]

0:15 Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar

0:23 University of Science and Technology Hisar harana I welcome you all to the

0:30 video lecture series of cyber security information security in this lecture we will study

0:38 about ethernet security which is basically used to secure wired local

0:45 area network and metropolitan area network first of all please see the

0:52 contents which I shall cover after covering brief introduction about ethernet security I shall take up

1:01 1 minute, 1 second ethernet threats which will be followed by ethernet Security Solutions in last I shall take up Max

1:11 SEC protocol for ethernet security so now let us

1:18 start ethernet has its roots in the 1970s in a radio experiment called Aloha

1:26 net developed under the direction of Norman Abramson at the University of

1:34 Hawaii at about same time gerox Corporation created a landan with a data

1:41 rate of 3 Mbps using a carrier sense multiple assess with Collision detection

1:49 protocol csma by CD in 1980 the 10 Mbps ethernet version

1:58 1.0 spe specification was jointly released by digital Equipment

2:05 Corporation Intel Corporation and Grox Corporation this was picked up by i e

2:15 and became the i e 802.3 standard in

2:21 1983 a chunk of data transmitted by ethernet over the wire is called a frame

2:30 on an Ethernet Network only a single node should be transmitting a frame at

2:37 any time if multiple systems are transmitting simultaneously a collision will

2:45 occur which can cause both signals to fail and require the systems to transmit

2:53 their frames again these properties are summarized under ethernet

3:00 designation as a carrier sense multiple assess with Collision detection CSM by

3:08 CD protocol figure shows a wired network using ethernet ethernet supports both
3:18 optical fiber and copper cabling yellow color in figure shows optical fiber and blue
color shows
3:28 twisted pair care CES in figure we can see Server form and pces connected to
3:36 fast ethernet 100 Mbps network using switches the ethernet standards comprise
3:46 several wiring and signaling variants of the OSI physical layer in use with
3:53 ethernet the original 10 base 5 ethernet uses coxl cable as a shared medium while
4:03 the newer ethernet variant use twisted pair and fiber optic links in
4:11 conjunction with hubs or switches over the course of its history
4:18 ethernet data transfer rates have been increased from the original 10 megabits
4:25 per second to the latest 100 gbits per second with 400 gabs per second expected
4:35 by early 2017 10 MVPs speed known as ethernet 100
4:42 Mbps known as fast ethernet 1,000 Mbps known as gigabit Ethernet 10,000 Mbps
4:52 speed as 10g ethernet and 100,000 Mbps as 100g
5:00 ethernet 400 GB speed is also expected in
5:08 2017 ethernet is the only survivor of the wired Lan War it is hard to find an
5:16 IP packet that has not passed over an Ethernet segment one important reason
5:23 for this is ethernet's Simplicity and ease of configuration
5:30 however ethernet has always been known to be an insecure technology recent
successful malware
5:39 attacks and the move toward cloud computing in data center demand that
5:46 attention should be paid to the security aspect of ethernet now we shall move on to
known
5:56 ethernet related security threats the key historical reason for the security
6:04 vulnerabilities of ethernet is that security has never been a major consideration in
its
6:12 design the basis for attack is gaining assess to the Target ethernet
6:19 segment the attackers may be an Insider with full assess right or may have found
6:27 an ethernet connection in a public public space before we describe the most
6:35 prominent methods for attacking ethernet segment let us see how an attacker May
utilize the network
6:43 assess techniques used by attackers are learning about the private Network
6:51 topology and the network traffic for use in attack gaining control over switch is
7:00 routers or servers in the landan eaves
7:06 dropping manipulating information disrupting the availability of the
7:15 network the most prominent methods for attacking ethernet segments
7:22 are network and system assess traffic confidentiality traffic
7:31 Integrity denial of service system

7:37 Security First of all let us see network and system assess type of
7:45 attacks assess to the network is a prerequisite for attacks and a necessity
7:52 for all types of attackers assess can be achieved by connecting equipment to the
network or
8:01 8 minutes, 1 second by gaining control of existing resources various methods
adopted for
8:08 gaining unauthorized access to network and system are unauthorized
8:16 joins connect to an Ethernet segment by gaining access to an unconnected port on
8:24 a switch it can be done by gaining physical access to the switch or gaining
8:32 access to a wall socket or removing the cable from a computer and plugging it
8:40 into another computer or plugging in a switch between the existing computer and
8:47 the socket unauthorized expansion of the
8:54 network the architecture of the ethernet allows users to access spend the network
9:01 9 minutes, 1 second by installing their own switches or wireless access points
which in turn
9:08 allows other people join the network vland
9:15 join if a switch listens for vland management protocol on host port a host
9:22 can act as a switch and join all vlans on some switches
9:29 the ports can be configured to not transmit VLAN management protocols but
9:36 they will still listen for these protocols an attacker can probe the
9:42 switch for hidden features like this vland tagging and
9:50 hoping an attacker can create ethernet frames that have a ven tag and thus
9:57 inject frames to V lanss to which they are not supposed to have access remote
access to the
10:06 landan access to an Ethernet segment can be achieved by gaining higher layer
10:13 access to a host on the segment for example by using social engineering and
10:20 to get a user at the Target Network to open a remote system administration
10:28 service which then connects to a host on the internet and enables the attackers to
access the
10:36 ethernet layer switch control as previously mentioned switches
10:44 are shipped with default or no passwords and the password can usually be
10:51 physically reset if an attacker gains control of a switch traffic can be
10:58 rerouted by switching links down claiming the STP route by Rising the
11:05 priority of the switch or dose selected links second type of attack methods are
11:14 traffic confidentiality traffic on the network can be useful in itself and also serve
11:23 the attackers in search of Targets in such type of attack
11:30 an attacker not only gains information being transmitted but also gets
authentication information like

11:39 passwords and network topology information that can be used for further
11:46 purposes third type of attacks are of traffic Integrity type it is done by modifying
traffic on
11:56 the network for example an attacker can imitate A bank's web server to a user
12:05 and imitate the user to the bank server and gain temporary control of the
12:12 users's bank account different ways to modify traffic on the network are ARP
12:21 and DHCP poisoning host capture traffic intended
12:28 for another host post just by sending an ARP message to the sender with the
12:35 intended receiver's IP address and the attacker's MAC address in a similar way a
host can
12:44 detect broadcast DHCP server request and raise the server to reply them first
12:53 upon success the attackers can assign a Gateway and DNS server to the Target
13:00 host along with its IP address and control the host traffic at
13:07 will man in the middle if an attacker can direct traffic to pass through his node and
that
13:17 traffic is not protected by an Integrity verification mechanism the attacker can
13:24 easily modify the traffic such attackers are called man in the middle
13:31 attack session hijacking ethernet is a stateless protocol but many higher level
protocols
13:41 create a session once a session is set up it is often assumed to be trusted and no
13:50 further traffic verification is made if an attacker can Eaves drop on or
13:59 otherwise gain enough information about an session the attacker can recreate the
14:06 session and act like one end point next type of ethernet attacks are
14:13 denial of service attack the attacker's motivation for dose attacks is not to gain
access to
14:23 data but to prevent its use the attacks can cause total loss of service or
14:31 degradation of service a dose attack can be performed in various
14:37 ways resource exhaust attack resource exhaust attack can Target the control
14:46 and management planes of a switch by sending frames that require additional
14:52 processing and handling protocol based dose
14:59 the STP that makes a tree out of a mesh network is designed to be
15:06 self-configuring an attacker that controls a node on the network can send
15:12 STP messages and pretend to be a switch the whole switching Network can be
15:20 brought to halt by flooding it with STP tcns or other STP control messages
15:30 is next we shall see solutions to Common ethernet attacks the security of ethernet
has
15:38 been improved by standard organizations equipment vendors and the
15:45 research Community traditionally ethernet lack of security has been

15:52 solved by defining any ethernet segment as unsecure and require iring it to be
15:59 placed inside a protected domain like placing it behind a firewall in a secure
16:07 building with trusted staff higher layer cryptographic
16:12 Solutions like IPC and TLs are used to solve the remaining
16:20 issues cryptography carries its own cost and thus cannot be considered a
16:27 universal Sol solution now let us see few existing Security
16:36 Solutions first of all let us see router based security methods in this Solution
16:44 One Central ethernet switch is replaced with an IP router the IP router
16:51 partitions ethernet Network into several segments each new segment is a separate
17:00 broadcast domain ARP STP VLAN and Mac address
17:07 table based attacks are no longer possible between the segments inside the
segment the same
17:16 attacks remain feasible unless each switch is replaced with a multi-port
17:23 router replacing a switch with a router will incur some cost
17:29 an IP router requires configuration such as address allocation and default route
17:39 configuration occasionally this can be automated as is the case for residential
17:45 Broadband where the topology is clear to the assess router the router also
17:53 prohibits easy Mobility a host may move in the ethernet Network nwor and keep its
IP and Mac
18:01 18 minutes, 1 second addresses as it is the MAC address tables in switches are
updated
18:09 automatically compared to an ethernet switch an IP router provides a considerable
amount of protection
18:18 against other users connected to the same router second method is assess control
18:26 method an attacker needs assess before being able to perform any attack
18:34 untrusted entities can be kept out by limiting assess to the network or requiring
18:42 authentication few methods of protection using assess control mechanism
18:49 are physical protection of the network network equipment can be located in
18:56 locked cabinets and rs and wiring installed inside walls to prevent unauthorized
19:04 assess segmentation and V Lans inside ethernet the i e
19:12 802.1q virtual L mechanism provides a way to limit broadcast and other traffic
19:20 to specific segments vlans are logically separate within the same physical
installation
19:28 and Define security domains inside one network individual
19:34 vlans each host on the ethernet can also be placed into its own VLAN using i e
19:44 82.1 a q inq double tagging or vendor
19:50 provided private vand switch configuration the PV Lan technique uses
19:58 switch configuration to isolate host and only let their traffic pass to one

20:04 promiscuous port typically connected to a router and to the
20:10 internet each host sees only itself and the host connected to the promiscuous
20:18 port and only a few VLAN IDs are needed at the trunks to indicate PV Lan traffic
20:27 access control list access control list security feature is not part of the ethernet
20:37 specification switch vendors have added various types of capabilities by
20:46 themselves the ethernet frame does not have many features for a simple ethernet
20:52 frame ACL the usable attributes are the sender's or
20:59 receiver's Mac address or The Ether type field access can be limited based on Mac
21:07 addresses but several service specific ACLs are commonly
21:14 implemented authentication based access control I E
21:21 802.1x port authentication supports several types of authentication credentials
21:29 such as a username and password pair or a certificate and corresponding private
21:37 key in i e 802.1x the client communicates with the
21:44 authentication switch and the switch verifies the authentication from the
21:51 database in authentication server as shown in figure 802 .1x uses extensible
22:02 Authentication Protocol EAP that has broad support for different
22:08 types of authentication methods and structures 802.1x authenticates a host at the
22:18 beginning of a session attaching the MAC address to a specific port in the
22:26 switch so this these were access control based Security Solutions which limit the
22:34 availability of targets to attackers now we shall see secure
22:40 protocols for ethernet security the targets can also be made harder to reach
22:47 by adding security features to protocols and such security Protocols are
22:54 implemented by following methods encryption and integrity
23:02 verification cryptography can solve integrity and confidentiality requirements i e
23:11 802.1 AE Max SEC forms encrypted connections between host and switches
23:21 protecting confidentiality and integrity of the content in the frame
23:29 deployment requires software installation and configuring authentication for each
participating
23:39 Network entity maxc provides protection against
23:45 Intruders to the network preventing reading and modification of data
23:52 frames we will see details of Max protocol later in this lecture
24:00 securing address resolution protocol ARP creates a major
24:07 vulnerability in the ethernet architecture information gained by DHCP snooping
can be used to prevent ARP
24:17 spoofing attacks by attaching Mac addresses to their corresponding IP
24:23 addresses and ports based on information gained from DHCP messages DHCP
snooping

24:34 suffers from a potential lack of scope as a single switch cannot see the
24:40 allocations made to host whose path to the DHCP server does not pass through this
24:49 switch last type of security methods are security monitoring based solution
24:58 in previous sections security techniques that are mostly proactive and once set
25:05 do not require active participation from external systems or human
25:13 interaction were described active Technologies enhance the protection of the
network by
25:21 implementing security monitoring based Solutions which are as explained
25:29 below ethernet firewall and deep packet inspection firewalls are used to limit
25:37 traffic between Network segments and can be considered more complex cases of
25:44 assess control list including stateful features current firewall products can
25:53 operate on all Network layers intrusion detection and prevention
26:01 26 minutes, 1 second systems intrusion detection system IDS and intrusion
prevention system IPS use
26:10 DPI to identify Network attacks usually from a signature library of known
26:18 attacks such systems are rarely useful for ethernet Network
26:25 protection planning configuration and Administration good Network
26:32 Administration and practices can considerably influence the aspect of an
26:38 Ethernet Network several of the Technical Solutions mentioned previously
26:45 require configuration and constant adjustment at the network topology
26:53 changes so these were techniques to secure ethernet Network now let us move
27:01 27 minutes, 1 second on to last part of this lecture that is maxc
27:07 protocol Max SEC that is Max security defined by i e
27:14 h2.1 AE standard allows authorized systems that attach to and interconnect
27:23 lens in a network they maintain confidentiality of transmitted data and
27:30 to take measures against frames transmitted or Modified by unauthorized
27:38 devices maxc protocol facilitates following maintenance of correct network
27:46 connectivity and services isolation of denial of service
27:53 attacks localization of any source of network communication to the land of
28:01 28 minutes, 1 second origin construction of public networks offering service to
unrelated or
28:08 possibly mutually suspicious customers using shared land
28:15 infrastructures secure communication between organizations using a landan for
28:23 transmission incremental and non disruptive deployment protecting the most
28:31 vulnerable Network components maxc operates in data part of
28:38 ethernet frame as shown in figure destination address and Source
28:44 address are same in mpdu or data part of

28:50 ethernet frame SEC tag is inserted before data and I see V is inserted
29:00 after data Max SEC comprises modification and additions to the max
29:08 service data unit msdu conveyed by each frame transmitted
29:16 by a user of the protocol the Mac security tag SE tag
29:24 conveys parameters that identify the protocol and identify the key to be used to
29:32 validate the received frame and provide replay protection these parameters are
TAG
29:41 control information TCI Association number a packet number
29:47 PN and secure Channel identifier SCI the secure data field conveys the US
29:58 user data encrypted if confidentiality is provided the icv ensures the Integrity
30:07 of the Mac destination address Mac Source address SE tag and user
30:17 data figure illustrates the transmission and reception of a frame by
30:24 maxc on transmission side the frame name is first assigned to an security
30:31 Association sa identified locally by its Association number the a is used to
30:40 identify the security Association key s AK and the next packet number PN for
30:49 that sa a the a sci and the PN encoded in the SE
30:58 tag are given to protect unone transmitter
31:04 side protect unit on transmitter side transmits icv and user secure data to
31:14 receiver on reip of mexc frame the a sci
31:21 and PN are extracted from the SE tag at receiver side
31:29 the A and seci are used to assign the frame to an sa and hence to identify the
31:38 security Association key s a k the validation function on receive
31:46 side of the current Cipher suit is presented with the sack
31:54 pnci the destination and source address of the frame together with the octets of
32:02 the SE tag secure data and icv if the Integrity of the frame has
32:10 been preserved and user data can be successfully decoded from the secure
32:17 data a valid indication and the octets of the user data are returned if the receive
frame
32:27 is Val replay protection is applied by checking that the received PN is not less than
32:36 the lowest acceptable PN for the sa if the check succeeds the parameter
32:43 of the frame unchanged from those transmitted are presented to the max SEC
32:50 receiver and lowest acceptable PN is updated and hence transmission is
32:58 successfully accepted so friends this was about
33:04 ethernet security finally summarizing we learned about introduction to ethernet
33:11 security which was followed by ethernet security threats and then ethernet
33:18 Security Solutions in the end we discussed working of very famous Max sex
33:27 security protocol hope the contents explained in this lecture were understandable
and

33:36 helpful hope to see you again in next lecture till then goodbye enjoy the day
33:43 thank you
33:46 [Music]
33:58 n
34:06 [Music]